HDOC-52

Previous Greater Element Using Pointers

Aim

To write a C program that reads a variable-length integer array and uses pointers to pass the array to a function previous_greater(). For each element, the function finds the nearest greater element on its left using a nested-loop approach; if none exists, it displays -1.

Algorithm

1. Read the size n of the array.

2. Declare a variable-length integer array arr[n] and read its elements.

3. Pass the address of the array to previous_greater() using a pointer.

4. For each element arr[i], search from i - 1 toward index 0.

5. The first element found that is greater than arr[i] is the nearest previous greater element.

6. If no greater element exists on the left, store/display -1.

7. Display all results in array order, separated by commas.

C Program

```
#include <stdio.h>

void previous_greater(int *arr, int n)
{
    int i, j, greater;

    printf("Previous greater elements: ");

    for (i = 0; i < n; i++)
    {
        greater = -1;

        for (j = i - 1; j >= 0; j--)
        {
            if (*(arr + j) > *(arr + i))
            {
                greater = *(arr + j);
                break;
            }
        }

        printf("%d", greater);
        if (i < n - 1)
            printf(", ");
    }
    printf("\n");
}
```

```
int main(void)
{
    int n, i;

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid array size.\n");
        return 0;
    }

    int arr[n];

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    previous_greater(arr, n);

    return 0;
}
```

Test Cases

Test Case	Input Array	Expected Output	Reason / Check
1	10, 4, 2, 20, 40, 12, 30	-1, 10, 4, -1, -1, 40, 40	Nearest greater value is searched only on the left.
2	1, 2, 3, 4, 5	-1, -1, -1, -1, -1	No earlier element is greater.
3	5, 4, 3, 2, 1	-1, 5, 4, 3, 2	Immediate previous element is greater each time.
4	4, 4, 5, 3	-1, -1, -1, 5	Equal values are not greater; strict > is used.

Sample Run

```
Enter the size of the array: 7
Enter 7 elements: 10 4 2 20 40 12 30
Previous greater elements: -1, 10, 4, -1, -1, 40, 40
```

Result

The program successfully finds and displays the nearest previous greater element for every array element using pointers and nested loops. It displays -1 when no greater element exists to the left.